# Guide to the Literature Supporting This Research

This appendix is derived from the literature review conducted for my thesis and presents the documents that supported its conceptual, methodological, and technical development. The same body of literature also contributed to the preparation of the scientific article arising from the thesis.

Its purpose is to provide readers with a clear and organized guide to the sources that were most useful during the research process. By presenting their main characteristics, topics, methods, tools, and applications, this appendix may help students, researchers, and professionals identify publications relevant to their own work and select documents they may wish to consult in greater depth.

The referenced documents are not reproduced or distributed with this appendix because I do not hold the authors' or publishers' permission to share copies of them. Instead, complete bibliographic references are provided so that readers can locate the original publications through their libraries, institutional repositories, publishers, or other authorized sources. All intellectual property rights remain with the respective authors and copyright holders.

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**Degree program:** Electrical Engineering

**Thesis title:** *Optimal Placement of FACTS Devices to Improve Voltage Stability in the 69 kV Subtransmission Network of the CNEL EP Guayas–Los Ríos Business Unit*

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## Methodology for Information Search

Following the methodology proposed by Torres-Carrión et al., the following research questions were formulated to delimit the scope of the study:

- RQ1: Foundations and classification of power-system stability.
- RQ2: Existing methods and techniques for assessing voltage stability.
- RQ3: Optimization methodologies and tools for the placement of FACTS devices.
- RQ4: Operation of power electronics in power-flow control.
- RQ5: Documented improvements and benefits in voltage stability after implementing FACTS devices.

To structure the body of knowledge, a conceptual mentefact was used. This graphical tool organizes the dimensions of thought associated with the central topic into supraordinate, isoordinate, infraordinate, and excluded concepts.

In addition, a thesaurus of terms was developed to establish hierarchies and synonym relationships among the keywords, thereby improving the precision of the literature search.

Finally, once the terminology had been defined, search strings were designed using Boolean operators adapted to each academic database. These search equations are based on three pillars: the problem (voltage stability), the strategy (FACTS/STATCOM), and the methodology (optimization and analysis).

1. SCOPUS

```
    TITLE-ABS-KEY (
    ( "voltage stability" OR "voltage profile" OR "voltage collapse" OR "voltage regulation" )
    AND
    ( "FACTS" OR "STATCOM" OR "Static Synchronous Compensator" OR "shunt compensation" )
    AND
    ( "optimal placement" OR "optimal allocation" OR "sizing" OR "optimization" OR "metaheuristic" OR "particle swarm optimization" OR "PSO" OR "genetic algorithm" OR "modal analysis" OR "sensitivity analysis" OR "PV curve" OR "QV curve" OR "DigSILENT" OR "PowerFactory" )
    )
    AND PUBYEAR > 2018

```

2. IEEE Xplore

```
("Voltage Stability" OR "Voltage Collapse" OR "Voltage Margin" OR "Reactive Power Control" OR "Weak Buses")
AND
("STATCOM" OR "Static Synchronous Compensator" OR "FACTS" OR "Flexible AC Transmission Systems" OR "Shunt Device")
AND
("Optimal Allocation" OR "Optimal Sizing" OR "Metaheuristic" OR "Particle Swarm Optimization" OR "DigSILENT" OR "Modal Analysis" OR "Sensitivity Analysis")

```

3. Google Scholar

```
"voltage stability" AND ("STATCOM" OR "FACTS") AND ("optimal placement" OR "allocation" OR "sizing") AND ("metaheuristic" OR "PowerFactory" OR "DigSILENT")

```

4. Springer Nature

```
"Voltage Stability" AND "Static Analysis" AND ("STATCOM" OR "FACTS") AND ("Optimal Placement" OR "Allocation" OR "Optimization Algorithm" OR "DigSILENT" OR "PowerFactory" OR "Smart Grid Simulation")

```

5. MDPI

```
("voltage stability" OR "voltage margin" OR "voltage collapse")
AND
("STATCOM" OR "FACTS" OR "shunt compensation")
AND
("optimal placement" OR "optimal allocation" OR "sizing" OR "metaheuristic" OR "particle swarm optimization" OR "power system simulation")

```

The search strings were used to review the relevant literature. The following table briefly describes the information reviewed and used to support and develop the project.

Artificial intelligence was used solely for proofreading grammar, spelling, and punctuation. Its use was limited to improving clarity and readability while preserving the authorship, originality, and technical rigor of the research.

| Source | Title | Author(s) | System | Tool | Technology | Focus | General topic | Specific topic | Year | Reference |
|---|---|---|---|---|---|---|---|---|---|---|
| Pearson | Power Electronics | Muhammad H. Rashid | Not applicable | Not specified | Not applicable | Fundamentals, devices, converters, and applications of power electronics | Power Electronics | Definition and conceptualization | 2017 | [1] |
| Escuela Superior Politécnica del Litoral | Static Voltage Stability Analysis Applied to the National Interconnected System | Adriano Alcívar Araujo, Dalton Maridueña Franco, Christian Larrosa Almendares | Ecuadorian National Interconnected System | UWPFLOW, MATLAB | Not applicable | Static voltage stability analysis of the National Interconnected System due to its importance in power-system operation | Voltage stability | Stability assessment methods | 2007 | [2] |
| Journal of Telecommunication, Electronic and Computer Engineering | Voltage Stability Assessment of Power System Network using QV and PV Modal Analysis | Ahmad Fateh Mohamad Nor, Marizan Sulaiman | IEEE 14-bus test system | MATLAB | Not applicable | Analysis of voltage instability in power systems using specialized strategies for a meshed network | Voltage stability | Stability assessment methods | 2016 | [3] |
| International Journal of Electrical Power and Energy Systems | Voltage vulnerability curves: Data-driven dynamic security assessment of voltage stability and system strength in modern power systems | Aleksandar Boricic, Marjan Popov | Specific IEEE test system | DigSILENT PowerFactory | Not applicable | A technique is used to dynamically assess voltage stability, enabling vulnerable areas of the power system to be identified and mitigated | Voltage stability | Stability assessment methods | 2025 | [4] |
| Universidad Técnica del Norte | Modeling of the EMELNORTE Subtransmission System and Its Integration with the Medium-Voltage Distribution System | Alicia Margarita Enríquez de la Torre | EMELNORTE power system | CYMDIST, DigSILENT PowerFactory | Not applicable | Modeling of the subtransmission system and its integration with the distribution system on a single platform | Voltage stability | Modeling and simulation | 2019 | [5] |
| Escuela Politécnica Nacional | Steady-State Voltage Stability Analysis of the Empresa Eléctrica Quito Subtransmission System | Antonio de Jesús Ortiz López | Empresa Eléctrica Quito (EEQ) subtransmission system | DigSILENT PowerFactory | Not applicable | Steady-state voltage stability study considering percentage load increases at load buses | Voltage stability | Stability assessment methods and DPL programming | 2012 | [6] |
| 2019 IEEE AFRICON | Steady-State Voltage Stability Assesment of the Kenyan High Voltage Transmission Network | Brighton Ombuki, Roy Orenge | Kenyan high-voltage transmission network | DigSILENT PowerFactory | Not applicable | Steady-state voltage stability assessment of the study system to identify critical lines and weak buses | Voltage stability | Stability assessment methods | 2019 | [7] |
| Escuela Politécnica Nacional | Power System Studies for the Electrical Engineering Program Using DIgSILENT PowerFactory 13.1 | Christian Wladimir Aguirre Cárdenas | Not specified | DigSILENT PowerFactory | Not applicable | Theoretical foundations of transmission lines, power flow, short circuits, protection, and small-signal and transient stability | Voltage stability | Modeling and simulation | 2008 | [8] |
| 2025 33rd Southern African Universities Power Engineering Conference (SAUPEC) | Improving voltage stability of a power system network using battery energy storage system (BESS) | Chukwuemeka E. Okafor, Azeez O. Olasoji, Komla Folly, D.T.O Oyedokun | IEEE 39-bus test system | DigSILENT PowerFactory | BESS | Implementation of a battery energy storage system as a compensator to improve voltage stability | Voltage stability | Stability assessment methods | 2025 | [9] |
| Universidad Politécnica Salesiana | Voltage Stability Improvement in a Distribution System Through Jacobian Modal Analysis for Power-Flow Control Considering Reactive-Power Compensators | Edgar Fabricio Zapata Veloz | IEEE 33- and 69-bus test systems | MATLAB | FACTS (SSSC - STATCOM) | Improvement of voltage stability through modal analysis in two test systems using a reactive-power support index | Voltage stability | Stability assessment methods | 2024 | [10] |
| Universidad Nacional Autónoma de México | Placement of D-FACTS in Transmission Systems for Power-Flow Control | Eliel Mendoza López | 4-bus system and Mexican Interconnected System | MATLAB | Distributed FACTS | Sensitivity analysis to determine suitable locations for distributed compensation devices | Voltage stability | Stability assessment methods | 2013 | [11] |

| Source | Title | Author(s) | System | Tool | Technology | Focus | General topic | Specific topic | Year | Reference |
|---|---|---|---|---|---|---|---|---|---|---|
| European Journal of Electrical Engineering and Computer Science | A comparative study of the influence of BESS and STATCOM on Post-Fault voltage recovery in microgrid | Fabliha Ahmed, Silvia Tasnim, Shourin Rahman Aura | IEEE 43-bus test system | DigSILENT PowerFactory | FACTS (STATCOM), BESS | A battery energy storage system and a STATCOM are used to improve post-fault voltage recovery in a weakly grid-connected microgrid | Voltage stability | Comparison of FACTS devices and BESS | 2022 | [12] |
| Universidad Politécnica Salesiana | Voltage Stability Study of Transmission Networks in the Northern Coastal Zone of Ecuador's National Interconnected System | Fernando Geovanny Carpio Marcillo, Armando Wladimir Romero Paucar | Ecuadorian National Interconnected System – Northern Coastal Zone | DigSILENT PowerFactorty | Contingencies | Voltage stability analysis of the study system and identification of vulnerable areas through contingency analysis | Voltage stability | Stability assessment methods | 2024 | [13] |
| Ingeniería y Desarrollo | FACTS Placement to Improve Voltage Stability | Henry Mauricio Rodríguez, Gladys Caicedo Delgado, John Edwin Candelo Becerra | IEEE 39-bus test system | Not specified | FACTS | Modeling and analysis of system operation, compensator placement, and comparison of different loading scenarios | Voltage stability | Stability assessment methods - Methodology | 2012 | [14] |
| Revista Ingenio | Voltage Equilibrium Analysis in a Subtransmission System Using QV-PV Curves and Modal Analysis | Hólger Santillán, Rogger Peña, Juan Morales | Subtransmission network of Manabí Province, Ecuador | DigSILENT PowerFactory | Not applicable | Analysis of the study system using voltage stability techniques to address unstable buses | Voltage stability | Stability assessment methods | 2021 | [15] |
| IEEE Power Engineering Society | Voltage Stability Assessment: Concepts, Practices and Tools | I. Dobson, T. Van Cutsem, C. Vournas, C.L. DeMarco, M. Venkatasubramanian, T. Overbye, C.A. Canizares | Not applicable | Not applicable | Not applicable | Voltage stability assessment covering basic concepts, common industry and research practices, and a detailed description of analytical tools | Voltage stability | Definition and conceptualization | 2002 | [16] |
| Información Tecnológica | Methods for Studying Voltage Stability in Power Systems | John E. Candelo, Gladys Caicedo, Ferley Castro | Not applicable | Not applicable | Not applicable | Review of voltage stability assessment methods in power systems, including a description and classification of the techniques | Voltage stability | Stability assessment methods | 2008 | [17] |
| Energy and Policy Research: Taylor and Francis | Voltage stability analysis of grid-connected wind farms with FACTS: Static and dynamic analysis | Kevin Zibran Heetun, Shady H. E. Abdel Aleem, Ahmed F. Zobaa | IEEE 14-bus test system | MATLAB - PSAT | FACTS (SVC, STATCOM) | Systematic approach in which static analysis evaluates voltage stability and dynamic analysis evaluates compensators | Voltage stability | Stability assessment methods | 2016 | [18] |
| SSRG International Journal of Electrical and Electronics Engineering | Synchrophasor Driven Voltage Stability Assessment Using Adaptive Deep Learning Based Tools on Temporal Ensembling and Data Augmentation | Kunal Samad, Arunkumar Patil, Katkar Satyapal, Diggikar Siddhant, Mohan N. Santosh | Not specified | Not specified | PMUs | Use of deep-learning models to predict voltage stability in power systems | Voltage stability | Deep-learning algorithms | 2024 | [19] |
| Ingeniería y Desarrollo | Effect of Series FACTS on Steady-State Voltage Stability | Luis Horacio Ramírez Borrero, John Edwin Cándelo Becerra, Gladys Nayiber Caicedo Delgado, Diego Burbano Jiménez | IEEE 300-bus test system | NEPLAN | Series FACTS | Description of voltage stability and series FACTS devices, together with a methodology for their optimal placement | Voltage stability | Stability assessment methods - Methodology | 2009 | [20] |
| IEEE TRANSACTIONS ON POWER DELIVERY | Voltage Stability-Based DG Placement in Distribution Networks | M. Ettehadi, H. Ghasemi, S. Vaez-Zadeh | IEEE 33-bus test system | Not specified | Distributed generation | DG placement based on voltage stability analysis and a reactive-power compensation method for prioritizing locations | Voltage stability | Stability assessment methods | 2013 | [21] |
| 2023 IEEE PES/IAS PowerAfrica | Improving Distribution Network Voltage Stability through Battery Energy Storage System | Mkhutazi Mditshwa, Ali Almaktoof, Yohan Darcy Mfoumboulou, Mohamed Tariq Kahn | IEEE 33-bus test system | DigSILENT PowerFactory | BESS | Use of BESS to improve voltage stability in distribution networks | Voltage stability | Alternative improvement techniques | 2023 | [22] |

| Source | Title | Author(s) | System | Tool | Technology | Focus | General topic | Specific topic | Year | Reference |
|---|---|---|---|---|---|---|---|---|---|---|
| IEEE Power India International Conference | PV/QV Curve based Optimal Placement of Static Var System in Power Network using DigSilent Power Factory | Neha Manjul, Mahiraj Singh Rawat | IEEE 14-bus test system | DigSILENT PowerFactory | FACTS (SVS) | Voltage stability analysis techniques for identifying the best location for SVS devices | Voltage stability | Stability assessment methods - FACTS devices | 2018 | [23] |
| Escuela Politécnica Nacional | Voltage Stability Analysis of the National Interconnected System | Nelson Victoriano Granda Gutiérrez | Ecuadorian National Interconnected System | UWPFLOW, MATLAB | Not applicable | Review and application of methodologies for voltage stability studies of Ecuador's National Interconnected System | Voltage stability | Stability assessment methods | 2006 | [24] |
| IEEE Transactions on Power Systems | Definition and Classification of Power System Stability – Revisited and Extended | Nikos Hatziargyriou, Jovica Milanovic, Claudia Rahmann, Venkataramana Ajjarapu, et al. | Not applicable | Not applicable | Not applicable | Expanded and updated definitions and classifications of fundamental stability terms considering the dynamic behavior of modern power systems | Voltage stability | Definition and conceptualization | 2021 | [25] |
| Scientific African | An easy method for simultaneously enhancing power system voltage and angle stability using STATCOM | Nnaemeka Sunday Ugwuanyi, Ogechi Akudo Nwogu, Innocent Onyebuchi Ozioko, Arthur Obiora Ekwue | Nigerian 50-bus power system | MATLAB - PSAT | FACTS (STATCOM) | Optimization of FACTS device placement to improve voltage and rotor-angle stability | Voltage stability | Stability assessment methods | 2024 | [26] |
| MDPI: energies | Voltage Profile and Sensitivity Analysis for a Grid Connected Solar, Wind and Small Hydro Hybrid System | Noah Serem, Lawrence K. Letting, Josiah Munda | IEEE 39-bus test system | Not specified | FACTS (D-STATCOM) | Voltage stability study of a network with high penetration of renewable energy sources | Voltage stability | Definition and conceptualization - Methodology | 2021 | [27] |
| IEEE TRANSACTIONS ON POWER SYSTEMS | Definition and Classification of Power System Stability | Prabha Kundur, John Paserba, Venkat Ajjarapu, Göran Andersson, Anjan Bose , Claudio Canizares, Nikos Hatziargyriou, David Hill, Alex Stankovic, Carson Taylor, Thierry Van Cutsem, Vijay Vittal | Not applicable | Not applicable | Not applicable | Definition and classification of power-system stability, focusing on instability modes, time scales, and associated disturbances | Voltage stability | Definition and conceptualization | 2004 | [28] |
| Revista Técnica Energía | Voltage Stability Analysis Using Modal Analysis | Wilmer Gamboa, Juan Plazarte | Ecuadorian National Interconnected System | Not specified | Not applicable | Methodology for identifying vulnerable buses and areas using voltage stability analysis techniques | Voltage stability | Stability assessment methods | 2007 | [29] |
| Springer | Voltage Stability Analysis of Power System | Yong Tang | Not applicable | Not specified | Not applicable | Theory, analysis, and enhancement of voltage stability in electric power systems | Voltage stability | Definition and conceptualization | 2021 | [30] |
| IJRET: International Journal of Research in Engineering and Technology | Voltage Collapse Mitigation by Reactive Power Compensation at the Load Side | V. Chayapathi, Sharath B., G. S. Anitha | WSCC 3-machine, 9-bus system | MATLAB | FACTS (SVC, STATCOM) | Description of power-system stability and an alternative for mitigating voltage collapse through reactive-power support | Voltage stability - FACTS | Stability assessment methods | 2013 | [31] |
| Editorial UTP | Introduction to Power System Stability | Alejandro Garces-Ruiz, Walter Gil-Gonzalez, Oscar Danilo Montoya Giraldo | Not applicable | Not specified | Not applicable | Modeling of electrical components, steady-state stability analysis methods, and transient-stability methodologies | Power-system stability | Definition and conceptualization | 2023 | [32] |
| Wiley | Power System Dynamics: Stability and Control | Jan Machowski, Zbigniew Lubosny, Janusz W. Bialek, James R. Bumby | Not applicable | Not specified | Not applicable | Principles, models, and controls associated with power-system dynamics and stability | Power-system stability | Definition and conceptualization | 2020 | [33] |

| Source | Title | Author(s) | System | Tool | Technology | Focus | General topic | Specific topic | Year | Reference |
|---|---|---|---|---|---|---|---|---|---|---|
| McGraw-Hill | Power System Stability and Control | Prabha Kundur | Not applicable | Not specified | Not applicable | Understanding, modeling, analysis, and mitigation of stability and control problems | Power-system stability | Definition and conceptualization | 1994 | [34] |
| MDPI: electronics | Comparison of Advanced Flexible Alternating Current Transmission System (FACTS) Devices with Conventional Technologies for Power System Stability Enhancement: An Updated Review | Andrea Carbonara, Sebastian Dambone Sessa, Angelo L'Abbate, Francesco Sanniti, Riccardo Chiumeo | Not applicable | Not specified | FACTS | Review of scientific and practical literature to assess the evolution of FACTS technologies | FACTS | Definition and conceptualization | 2024 | [35] |
| International Journal of Computational Intelligence and IoT | A Review on Optimal Placement of FACTS Devices | Arup Das, Subhojit Dawn, Sadhan Gope | Not applicable | Not specified | FACTS | Benefits, applications, classification, and techniques for the optimal placement of FACTS devices in power systems | FACTS | Definition and conceptualization | 2019 | [36] |
| IET Power Electronics | Static synchronous compensators (STATCOM): a review | B. Singh, R. Saha, A. Chandra, K. Al-Haddad | Not applicable | Not applicable | FACTS (STATCOM) | Comprehensive review of STATCOM technology and its research potential, classifying more than 300 research publications | FACTS | Definition and conceptualization | 2009 | [37] |
| Energy Reports | A comprehensive review of FACTS devices in modern power systems: Addressing power quality, optimal placement, and stability with renewable energy penetration | Ban H. Alajrash, Mohamed Salem, Mahmood Swadi, Tomonobu Senjyu, Mohamad Kamarol, Saad Motahhir | Not applicable | Not applicable | FACTS | Analysis of FACTS devices for improving power quality and maintaining stability in conventional power systems | FACTS | Definition and conceptualization | 2024 | [38] |
| 2020 6th IEEE International Energy Conference (ENERGYCON) | Losses Reduction in the Yendi-Bimbilla-Kete Krachi Power Sub-transmission Line Using One of Two Flexible AC Transmission System Devices | Erwin Normanyo, Philip Blewushie, Gregory Maabesog | Yendi–Bimbilla–Kete Krachi subtransmission line | NEPLAN | FACTS (SVC, TCSC) | Analysis of voltage stability and power losses, as well as the effect of implementing FACTS devices | FACTS | Definition and conceptualization - Benefits | 2020 | [39] |
| Springer: Journal of Modern Power Systems and Clean Energy | Overview of power electronics technology and applications in power generation transmission and distribution | J. M. Maza-Ortega, E. Acha, S. García, A. Gómez-Expósito | Not applicable | Not applicable | FACTS | Current status and future role of power-electronics technology in smart grids | FACTS | Definition and conceptualization | 2017 | [40] |
| Springer: PowerFactory Applications for Power System Analysis | Programming of Simplified Models of Flexible Alternating Current Transmission System (FACTS) Devices Using DIgSILENT Simulation Language | Jaime C. Cepeda, Esteban D. Agüero, Delia G. Colomé | WSCC 3-machine, 9-bus system | DigSILENT PowerFactory | FACTS (TCSC, SSSC, STATCOM, UPFC) | Methodology for implementing FACTS device models in DIgSILENT PowerFactory | FACTS | DSL programming (DigSILENT Simulation Language) | 2014 | [41] |
| Cogent Engineering: Taylor and Francis | Performance comparison of FACTS controllers for transmission pricing diminution | Kranthi Kiran Irinjila, Jaya Laxmi Askani, Kun Chen | IEEE 30-bus test system | Not specified | FACTS | Methodology for classifying and locating different types of FACTS devices | FACTS | Definition and conceptualization | 2018 | [42] |
| Universidad, Ciencia y Tecnología | Study and Modeling of FACTS Devices for Voltage and Reactive-Power Control in the National Power System | Mariam C. Rivera Ch., Omar E. Bermúdez R. | IEEE 14-bus test system - Sisnama Eléctrico Nacional | MATLAB - PSAT | FACTS (SVC) | Methodology for placing FACTS devices in power systems to control voltage and reactive power | FACTS | Definition and conceptualization - Benefits - Methodology | 2012 | [43] |

| Source | Title | Author(s) | System | Tool | Technology | Focus | General topic | Specific topic | Year | Reference |
|---|---|---|---|---|---|---|---|---|---|---|
| Heliyon | A comprehensive review of flexible alternating current transmission system (FACTS): Topologies,applications, optimal placement, and innovative models | Mehrdad Tarafdar Hagh, Mohammad Ali Jabbary Borhany, Kamran Taghizad-Tavana, Morteza Zare Oskouei | Not applicable | Not applicable | FACTS | Study of FACTS technologies, their application in power systems, and their associated challenges | FACTS | Definition and conceptualization | 2025 | [44] |
| 2014 International Conference on Energy Systems and Policies (ICESP) | Improving the voltage stability and performance of power networks using power electronics based FACTS controllers | S. Fahad Bin Shakil, Nusrat Husain, M. Daniyal Wasim, Shayan Junaid | Not applicable | Not applicable | FACTS (SVC, STATCOM) | Analysis of current power-grid trends in the application of FACTS devices to improve performance | FACTS | Definition and conceptualization | 2014 | [45] |
| MDPI: energies | Influence of FACTS Device Implementation on Performance of Distribution Network with Integrated Renewable Energy Sources | Filip Relic, Predrag Maric, Hrvoje Glavas, Ivica Petrovic | 20-kV rural distribution network | DigSILENT PowerFactory | FACTS (SVC) | Assessment of the influence of FACTS devices on the performance of a network with renewable energy sources using an algorithm that determines optimal placement | FACTS - Optimization | Definition and conceptualization - Benefits - Methodology | 2020 | [46] |
| IEEE TRANSACTIONS ON POWER SYSTEMS | Integration of a StatCom and Battery Energy Storage | Z. Yang, C. Shen, L. Zhang, M.L. Crow, S. Atcitty | Not applicable | Not specified | FACTS (STATCOM), BESS | Integrated STATCOM/BESS system for improving dynamic and transient stability | FACTS and BESS | Control systems and STATCOM hardware | 2001 | [47] |
| Springer | Advanced Smart Grid Functionalities Based on PowerFactory | Francisco Gonzalez-Longatt, José Luis Rueda Torres | Various | DigSILENT PowerFactory | Various | Smart-grid functionalities and a comprehensive set of guidelines for their implementation and assessment using DIgSILENT | Analysis tools | Modeling and simulation | 2018 | [48] |
| Springer | Modelling and Simulation of Power Electronic Converter Dominated Power Systems in PowerFactory | Francisco Gonzalez-Longatt, José Luis Rueda Torres | Various | DigSILENT PowerFactory | Various | Overview of power-electronic converters for simulation, from operating principles to implementation | Analysis tools | Modeling and simulation | 2021 | [49] |
| Springer | PowerFactory Applications for Power System Analysis | Francisco M. Gonzalez-Longatt, José Luis Rueda | Various | DigSILENT PowerFactory | Various | Comprehensive set of guidelines and applications for DIgSILENT PowerFactory | Analysis tools | Modeling and simulation | 2014 | [50] |
| MDPI: electronics | Combined Use of Python and DIgSILENT PowerFactory to Analyse Power Systems with Large Amounts of Variable Renewable Generation | Javier Jiménez-Ruiz,Andrés Honrubia-Escribano, Emilio Gómez-Lázaro | Proposed industrial network | DigSILENT PowerFactory - Python | Wind generation | Analysis of Python and DIgSILENT as tools for assessing the integration of wind power plants | Analysis tools | Modeling and simulation | 2024 | [51] |
| IEEE Access | A Multi-Objective Approach for Voltage Stability Enhancement and Loss Reduction Under PQV and P Buses Through Reconfiguration and Distributed Generation Allocation | Akhilesh Kumar Barnwal, Lokesh Kumar Yadav, Mitresh Kumar Verma | IEEE 33-bus test system | Not specified | Distributed generation, Network reconfiguration, Voltage control | Approach for improving voltage stability and minimizing losses through distributed-generation placement, network reconfiguration, and voltage control | Optimization | GWO - Multi-objective | 2022 | [52] |
| Ain Shams Engineering Journal | A novel optimal allocation of STATCOM to enhance voltage stability in power networks | Bilal H. Al-Majali , Ahmed F. Zobaa | IEEE 14-, 57-, and 118-bus test systems | Not specified | FACTS (STATCOM) | Optimization based on two control variables: STATCOM size and location | Optimization | MIDACO - Multi-objective | 2024 | [53] |
| Swarm Intelligence: Recent Advances, New Perspectives and Applications | Particle Swarm Optimization: A Powerful Technique for Solving Engineering Problems | Bruno Seixas Gomes de Almeida, Victor Coppo Leite | Not applicable | Not applicable | Not applicable | Description of the classical PSO algorithm, its variants, and hybrid heuristic-deterministic methods | Optimization | Definition and conceptualization | 2019 | [54] |

| Source | Title | Author(s) | System | Tool | Technology | Focus | General topic | Specific topic | Year | Reference |
|---|---|---|---|---|---|---|---|---|---|---|
| Universidad Politécnica Salesiana | Optimal Placement of FACTS Devices in Transmission Networks Using Intelligent Search | David Alberto Pérez Cruz | IEEE 9- and 14-bus test systems | MATLAB, GAMS | FACTS (SVS-SVR) | Use of a metaheuristic for optimal FACTS device placement in a transmission network | Optimization | HS - Cost minimization | 2017 | [55] |
| IOP Conference Series: Materials Science and Engineering | Optimal placement of FACTS devices using optimization techniques: A review | Dipesh Gaur, Lini Mathew | Not applicable | Not applicable | FACTS | Discussion and comparison of metaheuristic optimization methods for the location, type, and rating of FACTS devices | Optimization | Definition and conceptualization - Metaheuristics | 2018 | [56] |
| 2008 11th International Conference on Optimization of Electrical and Electronic Equipment | Improvement of voltage stability and reduce power system losses by optimal GA-based allocation of multi-type FACTS devices | H.R. Baghaee, M. Jannati, B. Vahidi, S.H. Hosseinian, H. Rastegar | IEEE 30-bus test system | Not specified | FACTS (SVC, TCSC, UPFC) | Allocation algorithm for FACTS devices considering cost and power losses | Optimization | GA - Multi-objective | 2008 | [57] |
| AIMS Energy | BESS based voltage stability improvement enhancing the optimal control of real and reactive power compensation | Habibullah Fedayi, Mikaeel Ahmadi, Abdul Basir Faiq, Naomitsu Urasaki, Tomonobu Senjyu | IEEE 30-bus test system | Not specified | BESS | Voltage stability analysis and continuous integration of a battery energy storage system (BESS) | Optimization | Output apparent-power minimization | 2022 | [58] |
| 2009 IEEE International Conference on Fuzzy Systems | A novel multi-level quantization scheme for discrete particle swarm optimization | Hwachang Song, Ryan B. Diolata, Young Hoon Joo | IEEE 37-bus test system | Not specified | Photovoltaic systems | Description of PSO from the conventional to the discrete model, followed by its application to a PV-system placement problem | Optimization | PSO - Definition and conceptualization | 2009 | [59] |
| 1997 IEEE International Conference on Systems, Man, and Cybernetics. Computational Cybernetics and Simulation | A discrete binary version of the particle swarm algorithm | J. Kennedy, R.C. Eberhart | Not applicable | Not applicable | Not applicable | Reformulation of the algorithm to operate with discrete binary variables | Optimization | PSO - Definition and conceptualization | 1997 | [60] |
| Hatun Yachay Wasi | Metaheuristics with Python | Jesús Espínola Gonzales, Ángel Cobo Ortega, Rocío Rocha Blanco | Not applicable | Not applicable | Not applicable | Potential use of metaheuristics for decision problems, including a description and implementation in Python | Optimization | Definition and conceptualization - Metaheuristics | 2022 | [61] |
| Wiley | Optimization of Power System Operation | Jizhong Zhu | Various | Not specified | Not applicable | Optimization techniques used in the operation of modern electric power systems | Optimization | Definition and conceptualization | 2015 | [62] |
| 2010 Eleventh Brazilian Symposium on Neural Networks | Applying a Discrete Particle Swarm Optimization Algorithm to Combinatorial Problems | Matheus Rosendo, Aurora Pozo | Not applicable | Not applicable | Not applicable | Hybrid PSO algorithm for discrete problems that retains the particle-velocity update concept | Optimization | PSO - Definition and conceptualization | 2010 | [63] |
| Springer: Journal of Modern Power Systems and Clean Energy | A review on applications of heuristic optimization algorithms for optimal power flow in modern power systems | Ming Niu, Can Wan, Zhao Xu | Not applicable | Not applicable | Not applicable | Recent applications of advanced heuristic optimization algorithms to optimal power-flow problems | Optimization | Definition and conceptualization - Metaheuristics | 2014 | [64] |
| International Journal of Computational Intelligence Systems | Optimizing FACTS Device Placement Using the Fata Morgana Algorithm: A Cost and Power Loss Minimization Approach in Uncertain Load Scenario-Based Systems | Mohammad Aljaidi, Pradeep Jangir, Sunilkumar P. Agrawal, Sundaram B. Pandya, Anil Parmar, Ali Fayez Alkoradees, Arpita, Aseel Smerat | IEEE 30-bus test system | MATLAB | FACTS, Wind generation | Algorithm for optimizing the location and size of FACTS devices | Optimization | FATA - Cost and loss minimization | 2025 | [65] |

| Source | Title | Author(s) | System | Tool | Technology | Focus | General topic | Specific topic | Year | Reference |
|---|---|---|---|---|---|---|---|---|---|---|
| Journal of Engineering and Applied Science | A review of FACTS device implementation in power systems using optimization techniques | Muddu Chethan, Ravi Kuppan | Not applicable | Not specified | FACTS | Assessment of methods for optimal FACTS device placement to improve system conditions | Optimization | Definition and conceptualization | 2024 | [66] |
| The MIT Press | Algorithms for Optimization | Mykel J. Kochenderfer, Tim A. Wheeler | Not applicable | Python - Julia | Not applicable | Optimization algorithms that explain how to explore the solution space to find the best solution | Optimization | Definition and conceptualization | 2019 | [67] |
| International Journal of Intelligent Engineering and Systems | Implementation Optimal Location of STATCOM on the IEEE New England Power System Grid (100 kV) | Najib Ababssi, El alami Semma, Azeddine Loulijat | IEEE 39-bus test system | MATLAB - PSAT | STATCOM | Location and sizing of a STATCOM to improve the voltage stability margin through a multi-objective formulation | Optimization | Multi-objective | 2022 | [68] |
| Universidad de Zaragoza | Multi-Objective Tabu Search for the Optimal Placement and Sizing of Distributed Generation and FACTS in Power Networks | Othon Aram Coronado de Koster | IEEE 300-bus test system | MATLAB - PSAT | Distributed generation and FACTS | Metaheuristic algorithm for solving multi-objective problems and comparing methodologies for selecting buses and lines | Optimization | TS - Multi-objective | 2020 | [69] |
| Scientififc Reports | Performance evaluation of STATCOM placement in real-world power system through DOANVSP algorithm for congestion management and stability improvement | Tayo Uthman Badrudeen, Funso K. Ariyo, Nnamdi Nwulu | Nigerian 28-bus power system | Not specified | FACTS (STATCOM) | Hybrid methodology for optimizing STATCOM location and size based on critical system points | Optimization | DOA - Voltage-stability-index minimization | 2025 | [70] |
| INTERNATIONAL JOURNAL OF RESEARCH AND SCIENTIFIC INNOVATION | Optimum Placement of Facts Devices on an Interconnected Power Systems Using Particle Swarm Optimisation Technique | Uduyok, Awajinwon Charles, Akaninyene B. Obot, Kufre M. Udofia | Nigerian 12-bus power system | MATLAB - PSAT | FACTS (STATCOM) | Optimal STATCOM placement using particle swarm optimization (PSO) | Optimization | PSO - Voltage-deviation minimization | 2025 | [71] |
| 2006 IEEE/PES Transmission and Distribution Conference and Exposition: Latin America | Optimal STATCOM Sizing and Placement Using Particle Swarn Optimization | Yamille del Valle, Ganesh Kumar Venayagamoorthy, Jean-Carlos Hernandez, Ronald G. Harley | Brazilian 45-bus power system | MATLAB - PSAT | FACTS (STATCOM) | Application of PSO to the optimal sizing and allocation of a static compensator | Optimization | PSO - Multi-objective | 2006 | [72] |
| IEEE Transactions on Evolutionary Computation | Particle Swarm Optimization: Basic Concepts, Variants and Applications in Power Systems | Yamille del Valle, Ganesh Kumar Venayagamoorthy, Salman Mohagheghi, Jean-Carlos Hernandez, Ronald G. Harley | Not applicable | Not applicable | Not applicable | Detailed description of basic PSO concepts and variants for power-system applications | Optimization | PSO - Definition and conceptualization | 2008 | [73] |
| MDPI: sustainability | Review of Metaheuristic Optimization Algorithms for Power Systems Problems | Ahmed M. Nassef, Mohammad Ali Abdelkareem, Hussein M. Maghrabie, Ahmad Baroutaji | Not applicable | Not applicable | Not applicable | Review of the applicability of several metaheuristic optimization algorithms to power-system challenges | Optimization | Definition and conceptualization - Metaheuristics | 2023 | [74] |
| 2016 IEEE 6th International Conference on Power Systems (ICPS) | Optimal location and sizing of STATCOM using Fuzzy-PSO approach | Bharat Singh Rana, Laxmi Srivastava | IEEE 30-bus test system | Not specified | FACTS (STATCOM) | Fuzzy-PSO algorithm for optimal STATCOM location and sizing | Optimization - FACTS | Fuzzy logic - PSO | 2016 | [75] |

| Source | Title | Author(s) | System | Tool | Technology | Focus | General topic | Specific topic | Year | Reference |
|---|---|---|---|---|---|---|---|---|---|---|
| IECON 2011: 37th Annual Conference of the IEEE Industrial Electronics Society | Particle swarm optimization for OPF with consideration of FACTS devices | H C Leung, Dylan Dah-Chuan Lu | IEEE 14-bus test system | Not specified | FACTS (UPFC) | PSO method for solving optimal power flow in power systems incorporating FACTS devices | Optimization - FACTS | PSO - Cost minimization | 2011 | [76] |
| MDPI: sustainability | Optimized FACTS Devices for Power System Enhancement: Applications and Solving Methods | Ismail Marouani, Tawfik Guesmi, Badr M. Alshammari, Khalid Alqunun, Ahmed S. Alshammari, Saleh Albadran, Hsan Hadj Abdallah, Salem Rahmani | Not applicable | Not applicable | FACTS | Comprehensive review of optimization techniques used in electrical engineering studies | Optimization - FACTS | Definition and conceptualization | 2023 | [77] |
| IEEE Access | Optimal Location of FACTS Devices in Order to Simultaneously Improving Transmission Losses and Stability Margin Using Artificial Bee Colony Algorithm | Mehrdad Ahmadi Kamarposhti, Hassan Shokouhandeh, Ilhami Colak, Banda Shahab S., Kei Eguchi | Two-area, four-machine system for low-frequency studies | Not specified | FACTS (SVC, TCSC, STATCOM, SSSC) | Optimization of FACTS device placement to improve power-system stability through a metaheuristic algorithm | Optimization - FACTS | ABC - Multi-objective | 2021 | [78] |
| Springer: Static Compensators (STATCOMs) in Power Systems | Optimal Placement and Sizing of STATCOM in Power Systems Using Heuristic Optimization Techniques | Reza Sirjani | IEEE 57- and 118-bus test systems | Not specified | FACTS (STATCOM) | Survey of literature focused on heuristic optimization techniques applied to STATCOM | Optimization - FACTS | GHS - Multi-objective - Methodology | 2015 | [79] |
| MDPI: energies | A Review on Optimization Objectives for Power System Operation Improvement Using FACTS Devices | Sohrab Mirsaeidi, Subash Devkota, Xiaojun Wang, Dimitrios Tzelepis, Ghulam Abbas, Ahmed Alshahir, Jinghan He | Not applicable | Not applicable | FACTS | Comprehensive review of existing proposals for improving power-system performance using FACTS devices | Optimization - FACTS | Definition and conceptualization | 2023 | [80] |
| Wiley | Power Generation, Operation, and Control | Allen J. Wood, Bruce F. Wollenberg, Gerald B. Sheblé | Not applicable | Not specified | Not applicable | Academic guide to planning, operating, and controlling electric generation and transmission systems | Electric power systems | Definition and conceptualization | 2013 | [81] |
| CRC Press: Taylor and Francis | Electric Energy Systems: Analysis and Operation | Antonio Gómez-Expósito, Antonio J. Conejo, Claudio A. Cañizares | Not applicable | Not specified | Not applicable | Description, understanding, and analysis of operation and planning problems in electric power systems | Electric power systems | Definition and conceptualization | 2020 | [82] |
| Divulgación y Actualidad Científica | Introduction to Electric Power Systems | Iván Matulic | Not applicable | Not applicable | Not applicable | Overview of electric power systems that convert energy into electrical form and transmit and distribute it | Electric power systems | Definition and conceptualization | 2003 | [83] |
| McGraw-Hill | Power System Analysis | John J. Grainger, William D. Stevenson Jr | Various | Not applicable | Not applicable | Introduction to the basic concepts and fundamentals of power-system analysis | Electric power systems | Definition and conceptualization | 2000 | [84] |
| CRC Press: Taylor and Francis | Electric Power Generation, Transmission, and Distribution | Leonard L. Grigsby | Not applicable | Not applicable | Not applicable | Presentation of electric power engineering integrating foundational knowledge across its major areas | Electric power systems | Definition and conceptualization | 2012 | [85] |
| Pearson | Electric Power Transmission and Distribution | S. Sivanagaraju, S. Satyanarayana | Not applicable | Not applicable | Not applicable | Guide to basic power-system concepts designed to progress gradually toward more complex topics | Electric power systems | Definition and conceptualization | 2010 | [86] |

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